

The Art of Gift-Giving with Limited Preference Data: How Fashion Recommender Systems Can Help

Sharareh Alipour*
sh.alipour@teias.institute
Tehran Institute for Advanced Studies
(TeIAS), Khatam University
Tehran, Iran
School of Computer Science, Institute
for Research in Fundamental Sciences
(IPM)
Tehran, Iran

Sina Elahimanesh*
sina.elahimanesh@sharif.edu
Sharif University of Technology
Tehran, Iran

Aliakbar Ghayouri Sales*
ghayoor@gmail.com
Sharif University of Technology
Tehran, Iran

Iman Mohammadi*
imanmohammadi@sharif.edu
Sharif University of Technology
Tehran, Iran

Parimehr Morassafar*
pmorassafar@gmail.com
Sharif University of Technology
Tehran, Iran

Seyed Parsa Neshaei*
seyed.neshaei@sharif.edu
Sharif University of Technology
Tehran, Iran

ABSTRACT

Gift shopping can be challenging due to the limited prior knowledge of the recipient's preferences, leading to after-purchase regret. The effectiveness of Fashion Recommender Systems (FRS) in the context of gift purchases with limited preference data remains under-explored. We considered a gift-buying scenario and conducted an experiment with 192 pairs of participants to compare FRS versus humans in recommending fashion gifts to buyers. We find both FRS and humans score >50% correctness in recommending the right gift, even without direct interaction with gift-givers or recipients. Although the buyers know the gift receivers directly, they lead to less accuracy. Additionally, we identify gender-based differences in the recommendations. We also embed our scenario into a smartphone application. Our findings investigate the potential of FRS in cold-start scenarios with limited data and unavailable human assistance while highlighting the risks of using FRS for in-store gift purchases.

CCS CONCEPTS

• **Information systems** → *Recommender systems*; • **Human-centered computing** → *Empirical studies in collaborative and social computing*.

KEYWORDS

Fashion Recommender Systems, User Evaluation, In-store Gift Purchase

*Contributed equally to this research.

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3651000>

ACM Reference Format:

Sharareh Alipour, Sina Elahimanesh, Aliakbar Ghayouri Sales, Iman Mohammadi, Parimehr Morassafar, and Seyed Parsa Neshaei. 2024. The Art of Gift-Giving with Limited Preference Data: How Fashion Recommender Systems Can Help. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3613905.3651000>

1 INTRODUCTION

Recommender systems have been extensively used in practice among people who aim to purchase gifts for their close friends, family members, or partners. More specifically, the usage of *Fashion Recommender Systems* (FRS) is on the rise, especially when incorporated into online shopping platforms (e.g., Amazon [29]). Online FRS work by creating a profile based on previous purchase preferences of users and then applying the knowledge derived from previous experience to select the best option as a gift. The more a user interacts with such shopping platforms, the more accurate the recommendation results will be. Recommender systems may also try to match the profiles of people together from social networks, as similar people tend to have similar preferences, and thus provide more accurate suggestions [28].

On the other hand, many users still prefer to shop in the *in-store* (i.e., *offline*) environments (e.g., department stores or shopping malls) in person. For such users, especially in stores that do not record the exact purchased clothing items along with the necessary metadata over time, only limited data is collected from previous purchases. This greatly limits how FRS can suggest the best possible gifts, as the literature has shown that recommender systems often experience noticeable performance hits when faced with only limited preference data (i.e., limited background data from users) [21]. Moreover, buyers may decide wrongly in purchasing items, possibly leading to an after-purchase regret [26]. While there has been previous work on enabling FRS to work with limited preference data [21], and thus provide a valid method for users to get

assistance in purchasing gifts, the literature lacks a thorough analysis of how FRS with limited data perform in the real-world scenario of buying a gift for a close friend, family member, or partner.

In the current work, we aim to explore the process of using FRS for in-store gift purchase scenarios by means of a user evaluation. We conducted interviews with 19 people, asking if they would use a recommender system for in-store scenarios. As most interviewees indicated the preference to use such a system, we found investigating in-store gift purchase settings to be beneficial. We define our research question as: *How effective are human- and computer-based recommenders in helping people purchase pieces of clothing as gifts for their close friends, family members, or partners, in the in-store and limited-data scenarios?*

To answer our research question, we obtained an FRS from previous works and then designed and conducted a user experiment acting as a simulation with 192 pairs of participants, who were all certified to be either family members, close friends, or partners. Each pair consisted of a gift buyer and a gift receiver. We asked the gift buyers to select, based on their opinion, the top three most preferred and least preferred clothes by the other, among a set of 12 clothing items given as pictures. By collecting the six selected options as the "limited data," the recommenders could get a grasp of the style the other person prefers. We then asked the gift buyer to select two options, in order of preference, for the other person, imagining if they are faced with choices in a real-world store among a set of clothing items. Then, we asked the gift receiver to select which one among the two options they would have preferred. This helped us acquire the ground truth for the gift purchase preferences. Based on the data gathered from the three most- and least-preferred options selected by the gift buyer, we used FRS as well as a group of human recommenders to select which one of the two options they would choose for the gift receiver. By comparing the results with the item chosen by the gift buyer and what was selected by the gift receiver, we found if consulting with a human- or computer-based recommender system has any positive or negative effect on choosing which item matches the most the preferences of the gift receiver.

The results from our comparison indicate that both using a) the FRS system, and b) a group of external human recommenders, help in choosing the final option selected by the gift receiver with higher accuracy than the gift buyer on average, and thus, can assist the gift buyer to purchase the better option if consulted with. Specifically, it is notable that there is no direct back-and-forth conversation or interaction between the gift buyer and the recommender; but still, choosing the output provided by the recommender, instead of the original item the gift buyer intended to purchase, increases the probability of the item being preferred by the gift receiver. In addition, we found differences among genders in the recommendation process. Among our human recommenders, we found females recommended the best clothing for female receivers and males recommended the best clothing items for male receivers. Also, the FRS we used had a noticeably higher performance when recommending clothes for males versus for females. Based on the promising results, we embed our method in a smartphone application prototype as well.

It is important to specify that we specifically chose *clothing* (fashion items) as our domain of recommendation investigation. This

is because we considered domains including clothing options in need of more qualitative and *objective* recommendation, relative to other more-*subjective* domains such as choosing digital devices. We considered *objective* domains as domains in which how individuals perceive their *emotions* with a product affects their purchase decision more compared to quantitative aspects. Additionally, it is important to note that we selected a certain type of fashion items to prevent any noise from possible different settings and situations in our study and make our measurement more reliable in this specific scenario.

The contributions from this paper are twofold. First, we provide a novel experimental design to simulate an in-store gift purchase situation and perform an analysis of how recommender systems can be used and how much they are effective in situations with limited customer data. We also discuss whether getting assistance from human- or computer-based recommenders helps the gift buyers choose the most preferred option for the gift receiver, using only limited user preference data as input. We also embed our method, due to the promising results we received, in a smartphone application prototype we designed for our exact scenario. Second, we highlight the risks of using FRS for in-store gift purchase situations, specifically regarding the differences in recommendation accuracy and correctness among genders. This work contributes to the human-computer interaction (HCI) research directions of exploring the challenges of using FRS in real-world scenarios, as well as comparing how performant FRS are when faced with limited background data on the user's preferences. We believe the results from this preliminary work can be used to enhance user-centric recommender systems working with limited data.

2 RELATED WORK

The practices for gift-giving and the associated social dimensions have long been explored in the literature [4, 11, 31]. In this work, we define a *gift* as a product or object purchased for giving to another person, while the receiver does not know they would receive the object beforehand. We consider *fashion* gifts (i.e., clothing items) as our domain of study, which is a well-known category of gift-giving in previous social research works [12, 19]. We specifically see previous works considering the differences in gift-buying behavior among gift receivers or gift buyers belonging to various gender, ethnicity, or age groups [7, 8, 14].

Fashion Recommender Systems. Today's recommender systems are capable of providing highly relevant suggestions to the users [1, 15]. FRS have been studied extensively in the past decade, with various approaches proposed for their development. Some researchers have focused on collaborative filtering methods, utilizing similarities between users and items to generate tailored suggestions [18, 25].

Other researchers have incorporated content-based techniques that consider the characteristics of clothing items, such as color, texture, and style, to provide tailored suggestions. For example, some works have used image processing methods to analyze the content of the input images [30]. Also, hybrid methods that combine the strengths of collaborative and content-based approaches have emerged [3]. Researchers have not only explored how to develop recommender systems but also the methods of how they can be

evaluated [22, 23], and the user experience of working with such systems [5, 16].

Human Recommenders. In previous works, human recommenders (e.g., as *experts*) have been employed to assist customers in selecting appropriate items and influencing purchase decisions [17]. However, research on the performance of human recommenders compared to FRS has been rare, especially in situations with few preference data.

Some studies have compared the effectiveness of FRS and human recommenders in various contexts, including but not limited to clothing selection. For example, [17] compared human recommenders with FRS in the domain of movie recommendations and found that the algorithm outperformed humans on average, but humans on average outperformed the system on certain prediction tasks.

On the other hand, [13] reported that human recommenders achieved better results when integrated into the loop for a collaborative filtering-based FRS. They incorporated human personality into their collaborative filtering framework to address the *cold-start* problem, in which the available data to start from, and base the recommendations on, is limited [13, 21, 24]. These studies, however, did not investigate the specific context of *gift selection* with limited data on the recipient's preferences.

Limitations in the Literature. The existing literature on FRS and human recommenders has mainly focused on the general fashion item selection process, overlooking the unique challenges associated with in-store gift selection in cold-start situations when only sparse data on users' preferences are available. While previous works have discussed cold-start scenarios for recommendation [6, 10, 32], the best practices for implementing them are under-explored in HCI research. Our work addresses this gap by examining the performance of FRS and human recommenders in the context of gift selection for an in-store setting with limited preference data available from the gift receiver.

Moreover, we explore the differences in recommendation strategies for male and female gift recipients (which, while explored in previous work [2, 20, 27], have not been thoroughly evaluated on in-store clothing gift purchase settings), providing a more comprehensive understanding of the factors influencing the success and correctness of FRS and human recommenders in such scenarios.

3 METHOD AND EXPERIMENT DESIGN

To find if consulting with recommender systems is deemed necessary in purchasing gifts for in-store scenarios, we first conducted preliminary interviews with 19 people (current or graduated students recruited from our university or personal contacts, as well as their families, each 10-15 minutes; 11 females and 8 males, average age = 34.95, STD = 19.10) who have had experience with purchasing clothes (i.e., confirmed to have purchased at least two clothes in the previous month) in-store. In the interview guidelines for our semi-structured interviews, we included the following four questions: 1) their previous experience and perception of in-store gift purchases, 2) if they are familiar with FRS, 3) if they would benefit from FRS for in-store purchases¹, and 4) what are the downsides

they associate with using FRS in this scenario. We took notes from the interviews, which two researchers coded together afterward, fixing any issues as they were brought up. We finally grouped the responses from the interviews, extracting clusters based on their opinion towards an in-store recommender system. The assignment of responses to the clusters was further confirmed to be correct by the investigation of two researchers. Thus, based on our interviews, we found that 15 people (78.95%) explicitly mentioned they would use an in-store recommender, adjusting its suggestion if they believe it does not make sense, and the rest (21.05%) mentioned they would use such a system given it has a high enough accuracy. No interviewee objected to having such a system, and all considered having access to such a system as beneficial. The results from our interviews signified that providing recommendations for in-store scenarios is well sought-after and worthy of investigation².

Thus, we designed an experiment simulating an in-store environment. For our computer-based FRS, we used a Vision Transformer (ViT) model [9]. ViT works by dividing an image into patches and processing them, similar to token sequences. We encoded all images using the model and then found the highest *dot score* between one of the source and one of the target images after calculating all pairwise dot scores. In other words, when ViT was given the image x and wanted to select the preferred image in the set V , it calculated a dot product between the ViT embeddings of x and each item $v \in V$, and selected the one with the maximum dot product³. We defined our experiment such that it can act as a viable simulation of the in-store gift purchase experience. While we could not completely emulate an in-store experience, we designed a simulation in which, similar to real-world stores, the *top* clothes to buy could be first chosen among a larger set, and then the buyer could make the final decision among a smaller set of filtered clothes. For each pair of people participating in the simulated experiment, we defined four main phases (see Figure 1 for a summary):

Phase 1: We show to the gift buyer one of the two sets of 12 pieces of clothing from a semi-diverse set of colors and styles, depending on the gender of the gift receiver. We used real-world images of clothes captured by phone cameras as opposed to getting the images from a dataset, as our goal was to simulate in-store shopping when high-quality pictures of the clothes are not necessarily available. After they carefully look at the 3*4 grid, we ask the gift buyer to indicate the top three most and least preferred clothes based on the preferences of the gift receiver. The selected photos act as inputs to the recommenders.

Phase 2: We ask the gift buyer to imagine they are in a clothing store and there is a set of 18 clothes (see Figure 1) to purchase as gift. We then ask them to identify the top two clothes (similar to Phase 1). The two clothes selected here would ideally be the top two

¹the interviewees think such a concept can be useful or not at all, before delving into details.

²We did not go in-depth in the interviews to collect design principles for an example of such recommender systems, as we considered our interviews to merely act as a preliminary investigation to see if exploring this research avenue is sought-after or not. We leave detailed interviews with experts to extract design considerations of such systems for future researchers.

³We did not fine-tune ViT to our data, because we wanted it not to have an advantage compared to the human recommenders in our experiment. We wanted both to generally follow the same approach according to the input data, that is, limited prior knowledge of the specific clothing items in our experiment. We also call for future work to incorporate more recent state-of-the-art recommendation engines into our experimental setup.

¹We acknowledge that there can be a potential bias in favor of FRS for in-store purchases based on the questions asked during the interviews. However, we first asked if



Figure 1: Overview of our experiment design, including its different phases. The picture grids on the left refer to the pictures used in Phase 1, and the ones on the right refer to Phases 2 to 4.

(or close to the top two) choices for the gift receiver (as the buyer knows the receiver personally). Thus, we assume that any doubts in selecting the best item as a gift are reduced to the problem of selecting the best item for the other person among the two selected ones in this phase.

Phase 3: We ask the gift buyer to select which of the two selected clothes in Phase 2 they would have preferred to purchase. The results would be an estimate of real-world in-store settings without consulting any recommenders. In addition, we use a) ViT and b) a group of human recommenders, give them the data from the previous phases, and ask them to select the clothing they would have recommended from the two selected options in Phase 2. Thus, the recommenders are only given limited data, similar to what they would also have access to in an in-store setting⁴.

Phase 4: We give the two selected clothes in Phase 2 to the gift receiver and ask them to select their preference. This data is designated as the ground truth; so, the recommenders can only be considered *helpful* if they can suggest what the gift receiver selected, rather than what the gift buyer selected previously.

For our experiment, we recruited 192 pairs of people among the students in our university and their families and friends (76 female gift receivers and 116 male gift receivers). We made sure that all pairs sufficiently knew each other's clothing preferences by checking if both had known each other for at least two years. Among the pairs with female gift receivers, 29 pairs were family members, and 47 pairs were close friends or partners, changing to 20 and 96 for male gift receivers. For the human recommenders,

⁴Ideally, in a real-world scenario, more details of the receiver are also revealed by the buyer to the recommender, for a better and more accurate recommendation. For the sake of simplicity in our experiment, as well as making the comparison between human recommenders and the ViT-based recommender fair, we only considered clothing preferences among a limited set of clothes as the transferred information. We expect the recommendation performance to improve further, with less possible bias, if more details of the buyers' profiles are revealed to the recommenders. We leave such improvements for future researchers.

we recruited five people in total. The main criterion for selecting the recommenders was previous experience with recommending clothes in informal, in-store scenarios. We purposefully did not recruit *expert* recommenders, as we aimed to explore how the recommendation is provided by people such as friends and family who can possibly accompany the buyer⁵.

To measure how much each recommender provides correct suggestions, we defined the correctness metric c_a as the proportion of all recommendations in which the gift receiver ranks the item that the recommender suggests, more than the other item⁶. The set of agents (a) to analyze the correctness metric for include a_1 = the ViT-based FRS, a_2 = majority vote of all humans, a_3 = majority vote of all females, a_4 = majority vote of all males, and a_5 = collective group decision of all humans⁷. This group setting simulates the situation where a group of people accompanying the gift buyer collectively provide purchase suggestions to them. The outcomes of a_5 are different from a_2 , as in a group setting, people can influence each other's decisions and reach a consensus which can be different from the majority opinion⁸.

Evaluating Gift Buyer Decisions. We noted that the gift buyers may not be completely familiar with what the gift receivers prefer, leading to adverse effects on our results. To investigate how our method behaves when the gift buyer is totally *familiar* with what the gift receiver prefers, we randomly chose 25% of the gift receivers and asked them to rate each item in the 3*4 clothes grid from -3 to +3, then normalized around their average per person. We then summed the scores given to the three best-selected clothes by the gift buyers and subtracted that of the three worst-selected clothes. This would ideally lead to a maximum of $3 \times 3 - 3 \times (-3) = 18$, so we discarded any data point below $18/2 = 9$. Finally, we found how correctly the agents recommended the items in this reduced subset. Reaching high correctness in this sub-experiment can affirm that the proposed setting works best when the gift buyer is familiar with the opinions of the gift receiver.

4 RESULTS

As can be seen in Table 1, in recommending clothes for female gift receivers, taking the majority vote of female recommenders was the most effective method, followed by male recommenders. The ViT FRS led to the lowest correctness score among the recommenders. For male gift receivers, taking the majority vote of male recommenders led to the highest correctness score, followed by group recommendation. Also notable is that the lowest correctness scores

⁵While we accept that using expert recommenders could have led to more accurate predictions, we only considered non-expert recommenders. This is in line with the common experience in real-world scenarios, where buyers do not necessarily have access to experts but can bring non-experts more easily along.

⁶We consider the entries in which the recommender's suggestion is same as that of the gift buyer, but different to that of the gift receiver, as entries excluded in decreasing c_a , because the recommender, in that case, has performed similarly to the gift buyer and the preferences indicated by them.

⁷For finding the group decisions, after each recommender provided their suggestions first, we organized a group setting in which all human recommenders attended and collectively reached a consensus for what to recommend in each entry.

⁸It should be noted that in this study, we did not conduct statistical significance tests, due to the nature of our data. We also did not conduct qualitative studies based on how the receivers perceived the gifts, thus not acknowledging enough the personal human *efforts* of selecting the gift.

Table 1: The correctness of each of the recommenders in our experiment, compared with the baseline of no recommender.

Agent / Correctness	Female gift receivers	Male gift receivers
FRS (ViT)	0.6471	0.7304
Male + Female Recommenders	0.7105	0.7241
Female Recommenders	0.7632	0.7241
Male Recommenders	0.7500	0.7586
Males + Females (group decision)	0.7368	0.7414
No Recommender	0.5000	0.5603

were achieved when no recommender was available⁹. Thus, consulting with any recommender increased the chance of purchasing the correct gift.

Additionally, our sub-experiment shows that all human recommenders suggested all the clothing items in the reduced subset correctly (correctness = 1.00), indicating a notable increase in the accuracy of our proposed experimental setup when the gift buyers know the preferences of the gift receivers generally well.

5 SMARTPHONE APPLICATION

As we got promising results based on our experiment, we decided to embed our method in a smartphone application (see Figure 2 for screenshots and the flow of our smartphone application). We develop our smartphone application using Flutter, enabling us to use a single codebase for both iOS and Android apps. The two main functionalities of our prototype smartphone app are as follows:

- **A) Buy a gift for a friend.** Twelve pictures are shown to the user to choose the items that their friend may prefer (page 2). Then, the user chooses three items that their friend dislikes (page 3). Pages 2 and 3 provide the necessary data source for the FRS to recommend clothing options upon. Then, the gift buyer should take two pictures of two items they want to select among them to get the recommendation of the ViT engine (pages 4 and 5). Additionally, the gift buyer can ask for a decision from human FRS anonymously in the hub of our app. This option is designated for when the gift buyer can wait a longer time to get peer recommendations.
- **B) Help others to recommend what they are going to buy.** On page 6, recommenders can see the preferences for each scenario submitted to the public system and help recommend one of the two items.

We propose our application as a prototype implementing our method in practice for future researchers.

6 DISCUSSION

In this study, we investigated the impact of providing recommendations from humans and FRS in the in-store gift purchase scenario. To do so, we designed a novel experiment to simulate the in-store gift purchase process. We used FRS from previous work, plus five human recommenders, providing suggestions without back-and-forth conversation with the gift buyers. For each recommender, we measured the correctness of the recommendations by finding for

which proportion of all entries the gift receiver ranked the item suggested by the recommender more than the other item. Our results indicate that even though the data points used by the recommender agents are limited¹⁰, the agents acted generally acceptable. The human recommenders led to the highest correctness score, with FRS following closely behind. Interestingly, we found that consulting with non-expert humans led to an improvement over consulting with no recommender. In fact, the buyers were the people who had knowledge about the recipient, but their own decision led to a lower correctness than when consulting with someone who does not know the recipient. Our findings show the mere act of consulting with an external agent, no matter if he knows the recipient, can improve the correctness score of buyers, compared to purchasing on their own. Finally, we uncover evidence of differences based on the gender groups and thus contribute to the research line of the influence of belonging to specific social groups on gift-buying behavior [7, 8, 14]. Each gender group in our study had a higher correctness score for gift receivers of the same group. Interestingly, the ViT FRS we used had a higher correctness score for male receivers compared to female receivers, which can possibly be related to either the distribution of training data used to originally train the ViT or the different complexities of recommending clothes for the two gender groups in this work. We thus believe our study can provide early insights for future researchers, be an influential prequel to motivate future work in this area, and stimulate conversations and ideas on how recommender systems can be used in in-store settings with limited collected data. Specifically, the findings from our study can be beneficial for HCI researchers designing user-centric recommender systems by considering the accuracy of their approaches among different scenarios and gender groups.

Limitations. We specifically chose fashion items as our domain of exploration on recommender systems, because we found a more qualitative recommendation necessary for domains such as fashion item recommender compared to other domains such as cars or smartphones, as the taste of individuals obviously influences the decisions more than certain quantitative metrics, such as car features or smartphone processing power. With that said, we find it important to also explore our findings in other recommendation domains as well in the future. Moreover, we only focused on certain types of clothing as we were going to measure our statistics without any noise of having different settings for the experiments. We call for future work to also consider other pieces of clothing in a similar setup to ours and measure any difference in results that may arise. Also, our experimental study is by no means a replacement for a

⁹For calculating this correctness score, we found the proportion of entries where the selection of buyer and preference of receiver matched.

¹⁰As the data contains a maximum of six liked and disliked clothing items for each gift receiver.

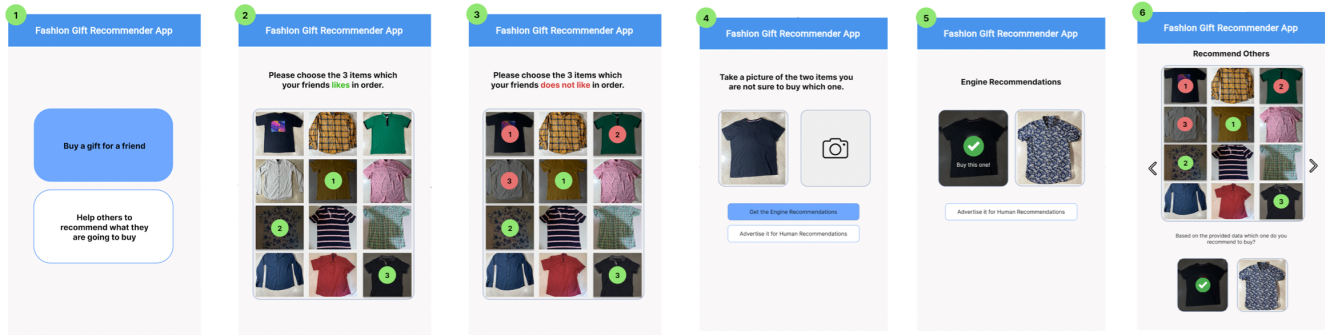


Figure 2: Overview of our smartphone app, developed as a prototype embedding our method.

real-world scenario, and thus, we call for future researchers to assess the effectiveness of our recommender agents in a real in-store scenario as opposed to a controlled limited experiment. In addition, the participants were only sampled from a certain region, and thus, the recommendation results may not be well generalizable. Moreover, the selected items, while illustrating as much similarity as possible to an in-store context, were still limited in number and style. We also call for future work to consider other metrics for evaluating the in-store recommendation helpfulness as well.

Regarding the FRS, while we used a ViT model to suggest clothing items, other state-of-the-art FRS algorithms, which we did not include in our study design, may lead to more accurate recommendations, acting better than the human agents even with limited prior data on the preferences of gift buyers. We expect future researchers to try replicating the study using other state-of-the-art FRS and report if any difference is observed. Moreover, due to the nature of the data (true/false correctness scores), we did not conduct statistical comparison tests across groups. We call for future studies to conduct more in-depth analysis of similarly collected data, accompanied by necessary robust statistical comparisons and tests, mitigating the over-reliance on the *correctness* score we measured, to bolster the study's empirical foundation. We also agree that gift-buying can encompass a wide range of challenges (e.g., clothing brands, changing trends over time, price of clothing items, or suitability for different occasions), which we did not consider separately in our work, and leave them for future researchers. We also did not conduct qualitative studies based on how the receivers perceived the gifts, as well as to see how the personal experience of buyers and receivers and the AI-mediated communication affected the perceptions of the participants. Finally, although we developed a smartphone app as a prototype embedding our method, we did not evaluate how effective such a design is among users. We call for future work to improve our smartphone app prototype and evaluate it in a study to confirm the specifics of the design of our application.

7 CONCLUSION

In this study, we designed a novel experiment simulating the situation in which gift buyers purchase clothes in-store. By analyzing the proportion of gift purchases where the gift receiver preferred the gift to the other item, thus leading to a *correct* recommendation,

we found consulting with recommenders has a positive effect on the purchases. We also found human recommenders generate more correct suggestions than the computer-based agent we used. We also investigated differences and emphasized the risks associated with using FRS and human recommenders in the in-store contexts regarding the gender of the gift receiver and the human recommenders. This study provides HCI insights into how recommender systems can be used in cold-start situations, where only limited data is available, and indicates the potential risks of using recommenders in in-store scenarios. We believe our study could be useful for future HCI and human-AI interaction researchers designing or implementing user-centric recommender systems for cold-start scenarios.

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